

#	Sr No	Unit Number	Topic	Status (P-Practical)					
	1	Unit 1	Overview: MPA, SPA, SSR, SSG	P1					
	2	Unit 1	Responsive Design, Atomic Design Methodology	Not covered					
	3	Unit 1	Overview: UI Component Libraries	Not covered	All uncovered topics are either overviews or theoretical concepts, which can be covered through an Expert Talk, Webinar, or Workshop.				
	4	Unit 1	Component Architecture	P1					
	5	Unit 1	State Management: Reactive Patterns	P2					
	6	Unit 1	Routing Concepts	P2					
	7	Unit 1	API Integration and Consumption	P3					
	8	Unit 2	Rendering Lifecycle	P3					
	9	Unit 2	Lazy Code Loading	P8					
	10	Unit 2	WCAG Principles	Not covered					
	11	Unit 2	Overview: PWA	Not covered					
	12	Unit 2	Performance Optimization	P8					
	13	Unit 2	Frontend Security (XSS, CSRF, Token Handling)	P7					
	14	Unit 2	Automated Testing (Application + Security level)	P12					
	15	Unit 3	Richardson Maturity Model	P4 (Assignment)					
	16	Unit 3	Middleware and Request Lifecycle	P4					
	17	Unit 3	RESTful API Design	P4					
	18	Unit 3	Data Validation Layers and Error Handling	P5, P7					
	19	Unit 3	Schema Enforcement	P5					
	20	Unit 3	In-memory Caching Strategies	P9					
	21	Unit 3	Secure Storage Principles	P7					
	22	Unit 3	Database Migration Principles	Not covered					
	23	Unit 3	Overview: Data Encryption	Not covered					
	24	Unit 4	Monolithic vs Microservices Architecture	P10 (Assignment)					
	25	Unit 4	Connection Pooling and Query Caching	P9					
	26	Unit 4	Horizontal vs Vertical Scaling	Not covered					
	27	Unit 4	Stateless Services	P11					
	28	Unit 4	Load Balancing and Fault Tolerance	Not covered					
	29	Unit 4	CDN Fundamentals	Not covered					

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	30	Unit 4	Asynchronous Processing and Message Queues	P10					
	31	Unit 4	Performance Load Testing (RPS, Latency, Error Rate)	P9 (Assignment)					
	32	Unit 4	CPU/Memory Profiling, Energy-Efficient Web Systems	Not covered					
	33	Unit 4	Site Reliability Engineering	Not covered					
	34	Unit 5	Overview: Cloud Computing, Cloud Native Systems	Not covered					
	35	Unit 5	Cloud Service Models Serverless, IaaS, PaaS	P11 (Assignment)					
	36	Unit 5	Containerization and OCI Standards	P11					
	37	Unit 5	Docker Fundamentals and Dockerfile	P11					
	38	Unit 5	CI/CD Pipeline	P12					
	39	Unit 5	Overview: Resource Limits Configuration	Not covered					
	40	Unit 5	Deployment Strategies Blue-Green, Rolling Updates	P12					
	41	Unit 6	Introduction: Threat Model	P7 (Assignment)					
	42	Unit 6	Advanced Authentication and Authorization	P7					
	43	Unit 6	Overview: HTTPS and TLS	P7 (Assignment)					
	44	Unit 6	Secure Coding: Introduction and Practices	P7					
	45	Unit 6	AI Service Integration and Prompt Engineering	P13					
	46	Unit 6	AI-assisted Customized Designing	P13 (Assignment)					
	47	Unit 6	Introduction: Responsible AI	P13					